Large Retail Bank – End-to-End Lending & Compliance Transformation

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Executive Summary

Led enterprise-scale transformation of a digital lending platform for a large retail bank, modernizing onboarding, underwriting, and compliance using a cloud-native, event-driven architecture.

Business Challenges

• Legacy monolithic systems delaying loan approvals
• Manual KYC and AML processes increasing turnaround time
• High customer drop-offs during onboarding
• No real-time fraud detection or risk scoring
• Fragmented systems across lending, compliance, and core banking

Solution Architecture

Designed and implemented a modern digital lending platform with:

• Event-Driven Microservices Architecture
  - Kafka-based asynchronous communication
  - Decoupled services for scalability and resilience
• Hybrid Multi-Cloud Strategy
  - Azure (primary platform)
  - GCP (AI/ML workloads and analytics)
• AI-Driven Decisioning
  - Real-time credit scoring
  - Fraud detection models
  - Automated underwriting
• Compliance Integration
  - Fenergo (KYC / onboarding)
  - Actimize (AML / transaction monitoring)
  - Sanctions and PEP screening
• Core Banking Integration
  - Finacle APIs for loan lifecycle

End-to-End Reference Architecture

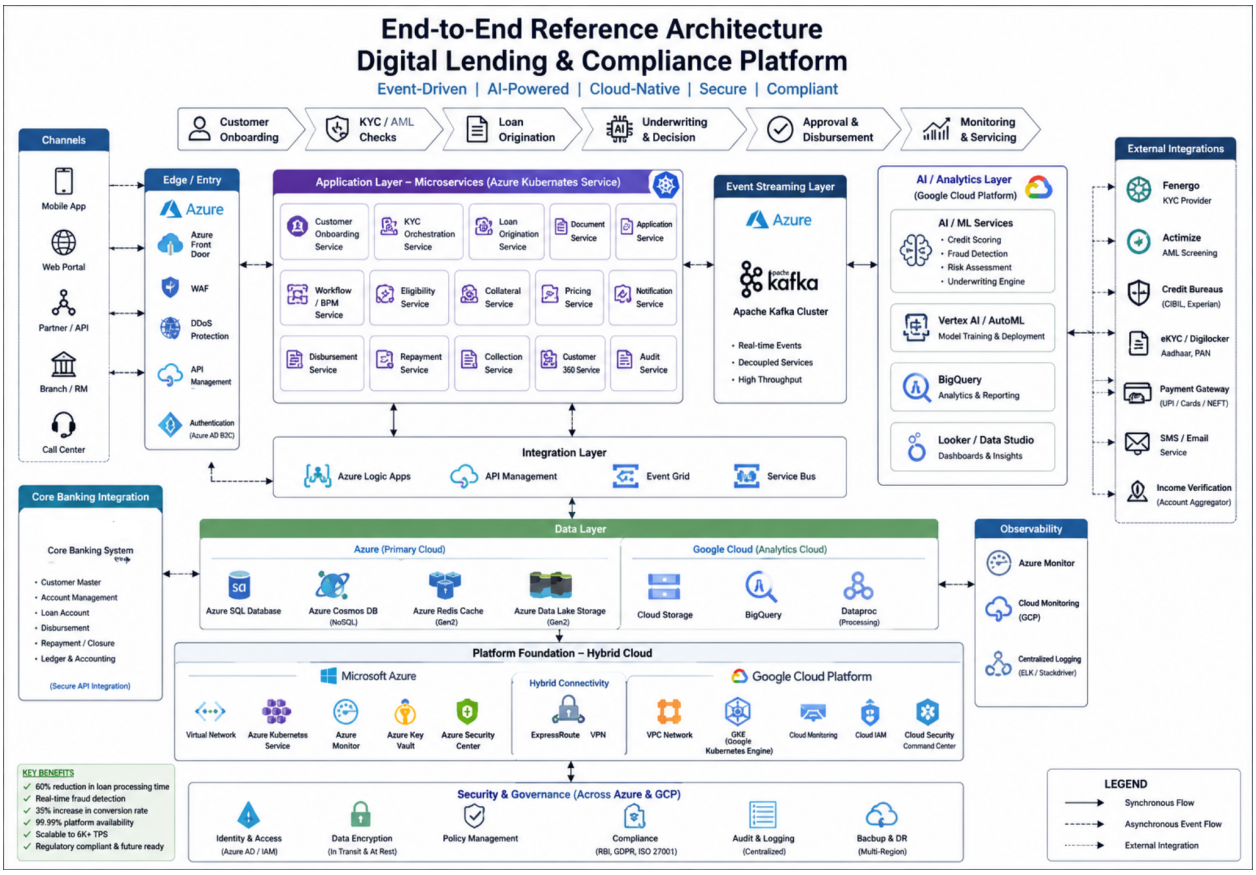

Figure: Enterprise Digital Lending Architecture (Event-Driven, AI-Powered, Multi-Cloud)

End-to-End Flow

1. Customer onboarding via Mobile/Web
2. API Gateway routes requests securely
3. Microservices handle onboarding + loan processing
4. Events published via Kafka
5. AI models evaluate risk + fraud in real-time
6. KYC/AML checks via Fenergo & Actimize
7. Approved loans pushed to Finacle
8. Data stored in distributed data platform

Business Impact

↓ 60% loan processing time
↑ 35% conversion
↓ 70% manual effort
99.99% availability
6K TPS throughput